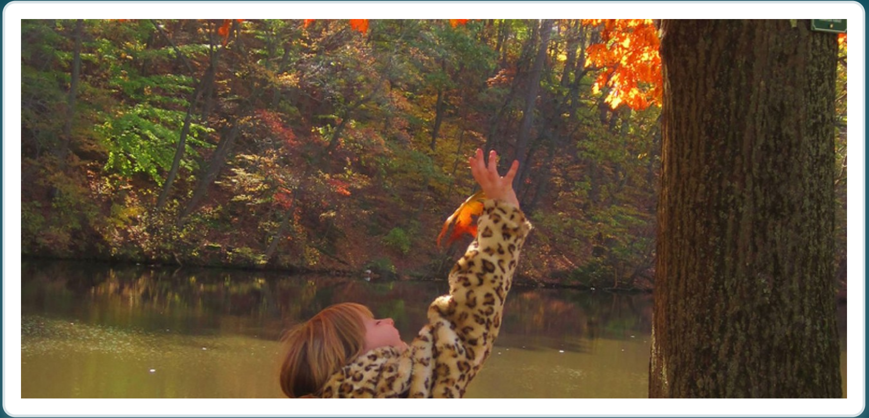
CHAPTER 17

The Endocrine System



Chemical control of the body through hormones.

Anatomy & Physiology · Broward-Miami Health Institute

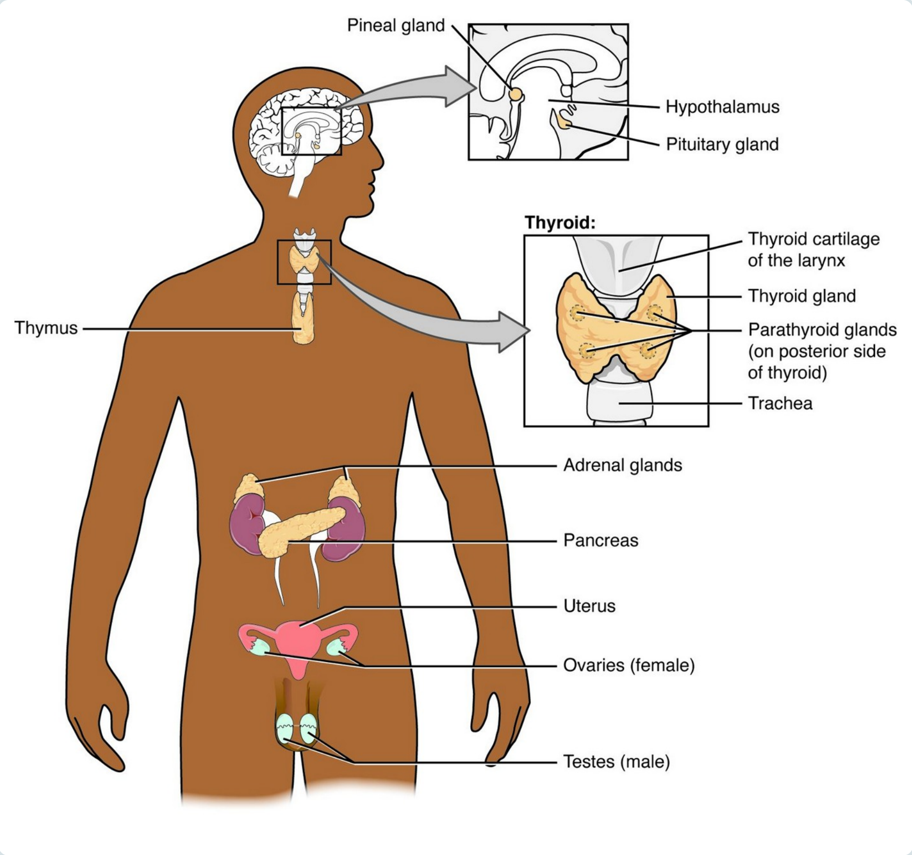
FACULTY EDITION

CHAPTER 17

Learning Objectives

After studying this chapter, students will be able to:

1. Identify the contributions of the endocrine system to homeostasis
2. Discuss the chemical composition of hormones and the mechanisms of hormone action
3. Discuss the hormonal regulation of the reproductive system
4. Explain the role of the pancreatic endocrine cells in the regulation of blood glucose
5. Identify the hormones released by the heart, kidneys, and other organs with secondary endocrine functions
6. Discuss several common diseases associated with endocrine system dysfunction



The major endocrine glands of the body.

17.1

SECTION

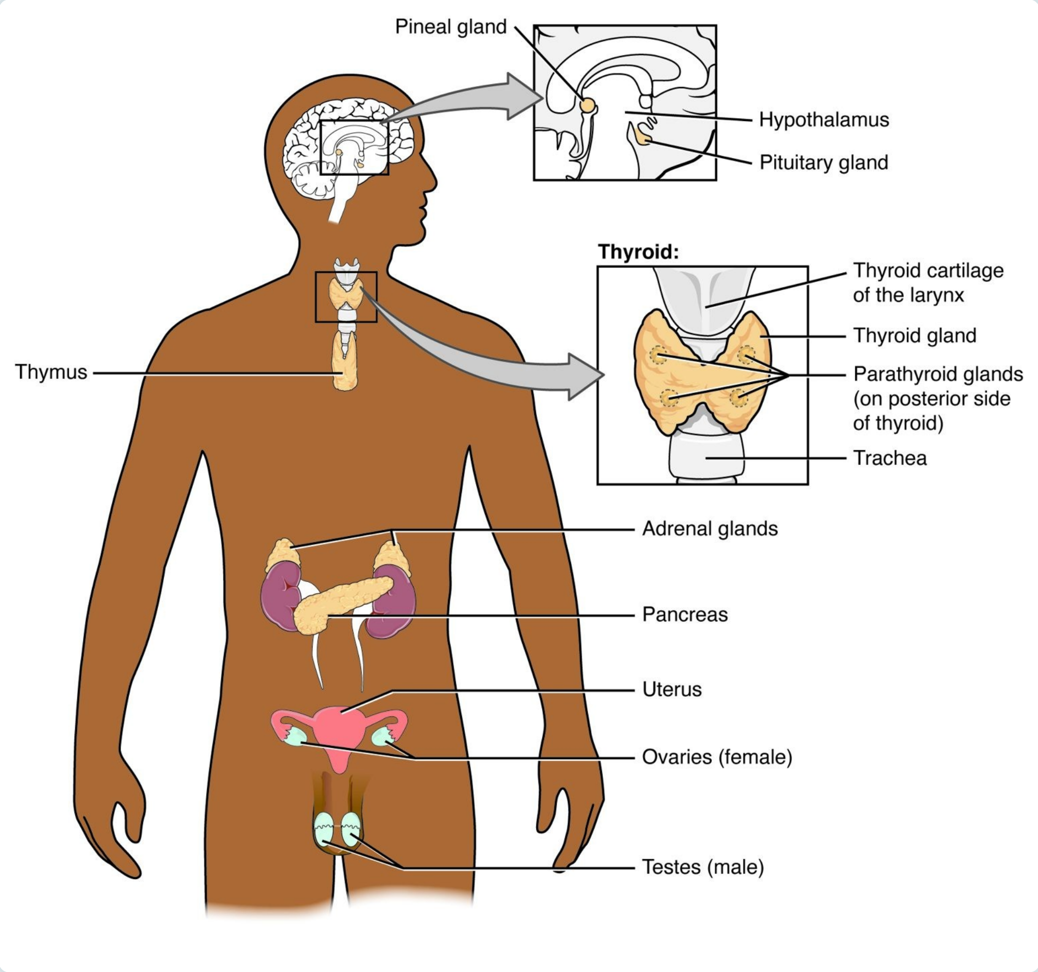
An Overview of the Endocrine System

The endocrine system — chemical control through hormones.

SECTION 17.1

The Endocrine Glands

- Glands secrete hormones into the blood
- Hormones travel to distant targets
- Control slow, widespread processes
- Work alongside the nervous system



The major endocrine glands

17.2

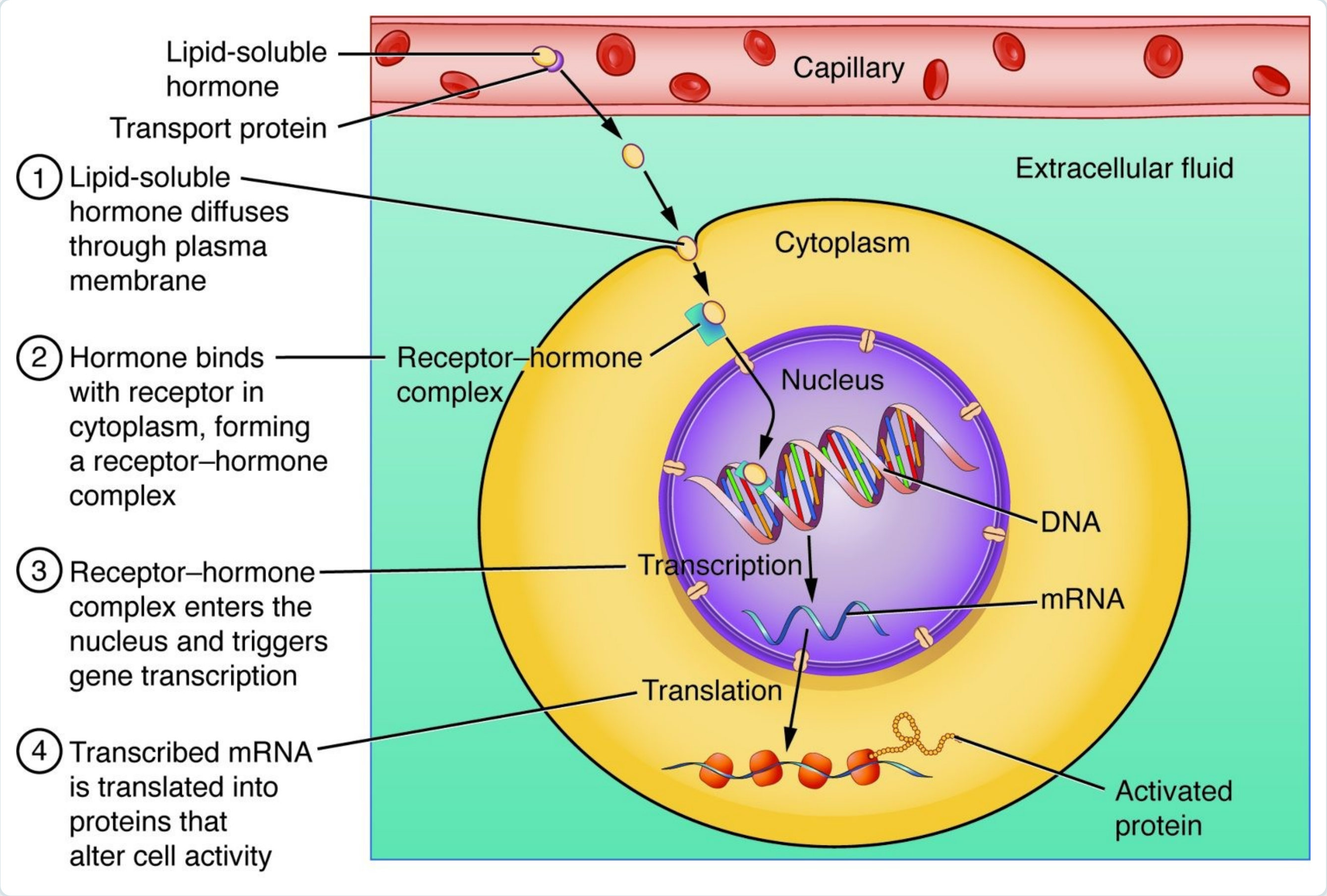
SECTION

Hormones

What hormones are and how they act on target cells.

Steroid Hormone Action

- Lipid-soluble hormones enter the cell
- They pass through the membrane
- Bind receptors inside
- Switch genes on or off

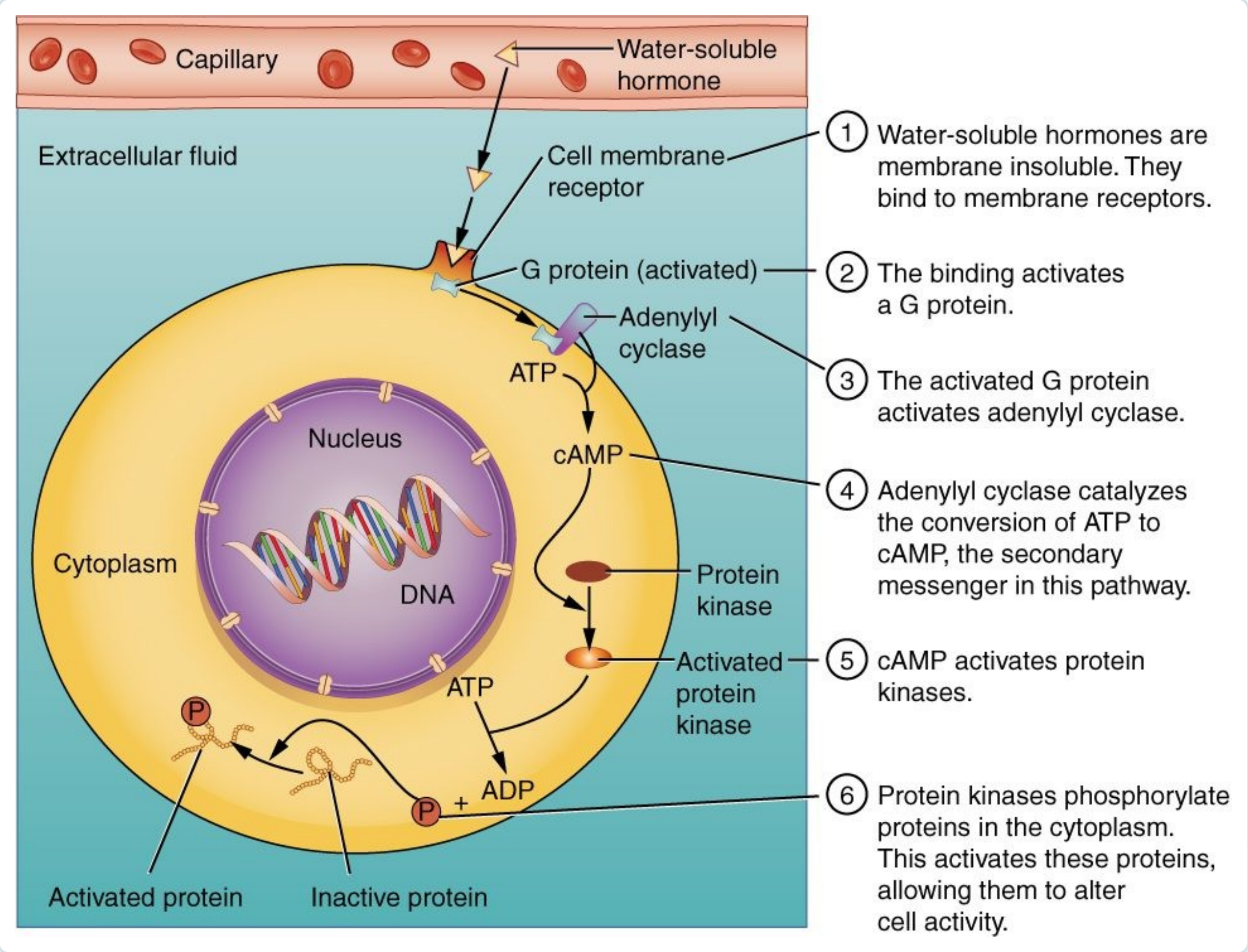


How a lipid-soluble (steroid) hormone acts

SECTION 17.2

Water-Soluble Hormone Action

- Water-soluble hormones can't enter the cell
- They bind receptors on the surface
- Trigger a second messenger (cAMP)
- Produce rapid responses

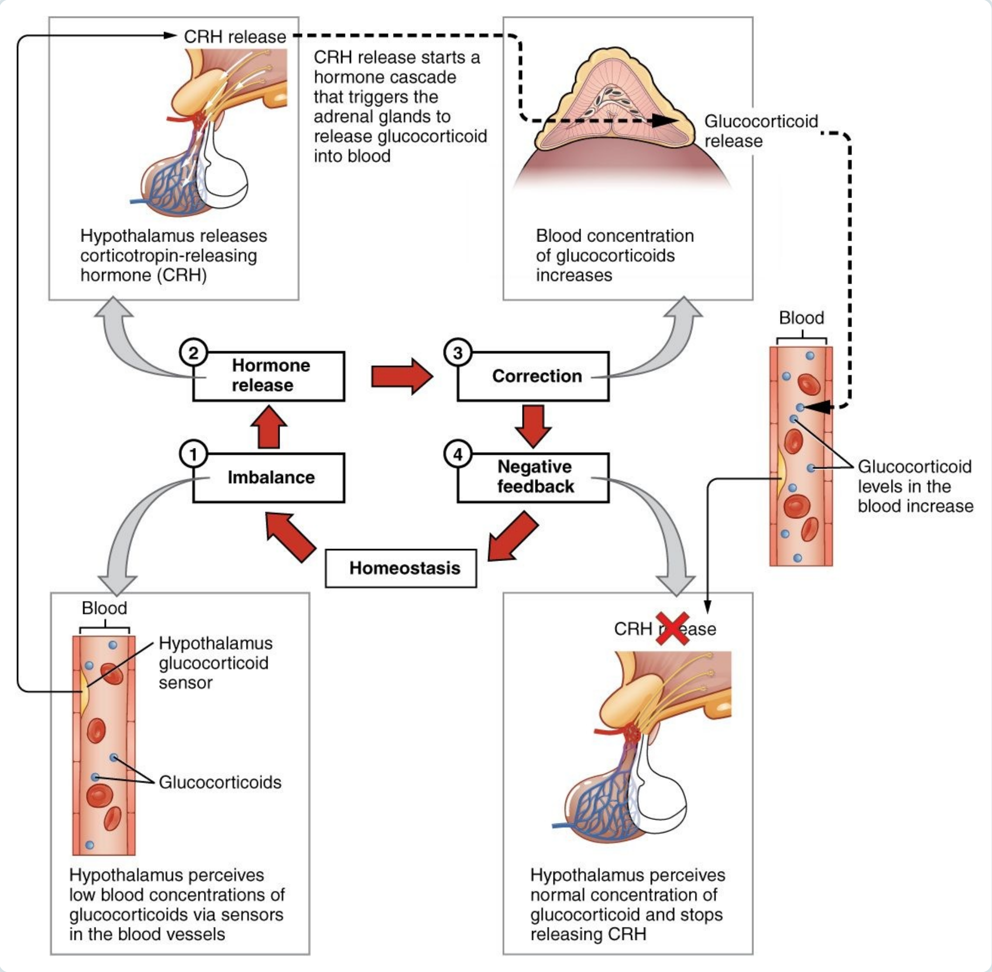


How a water-soluble hormone acts

SECTION 17.2

Negative Feedback

- Most hormones are feedback-controlled
- A change triggers a correcting response
- The response shuts off the signal
- Keeps levels in a stable range



Negative feedback control of hormones

17.3

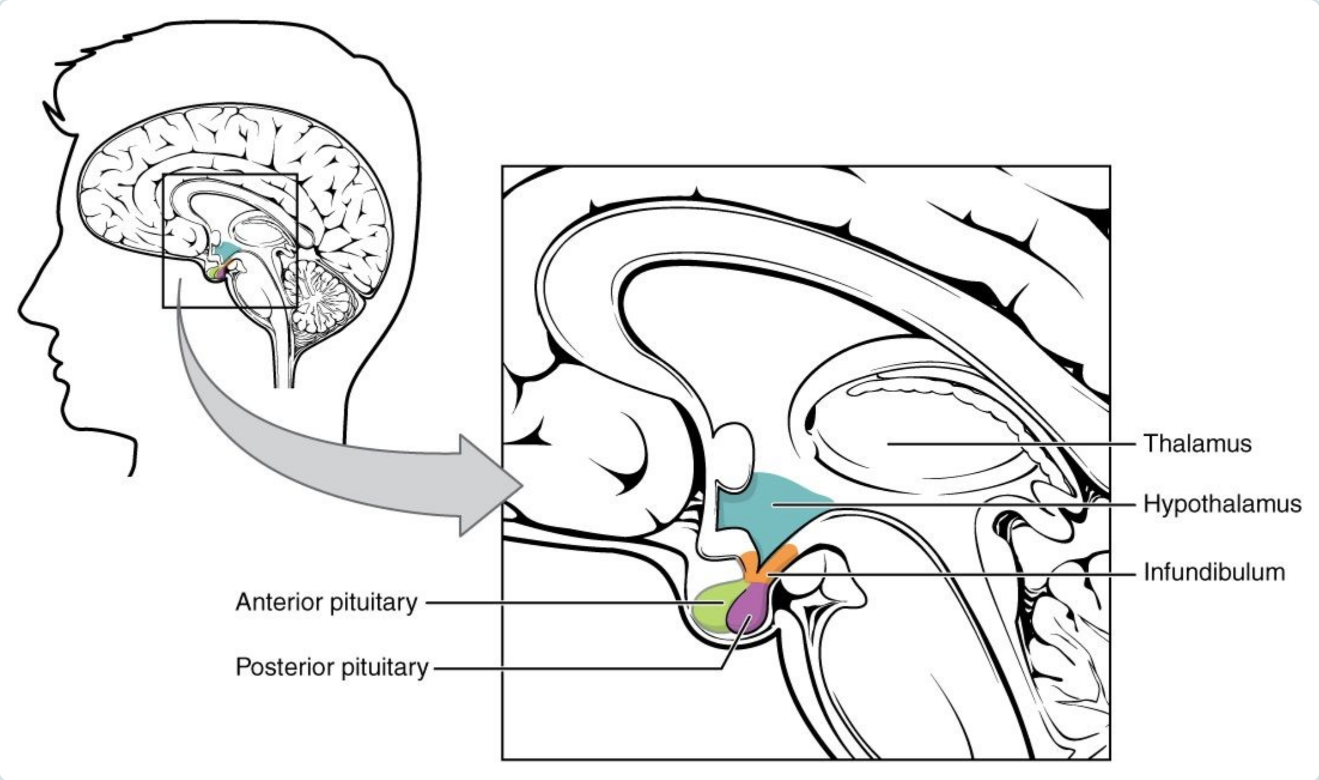
SECTION

The Pituitary Gland and Hypothalamus

The pituitary and hypothalamus — the hormonal command center.

SECTION 17.3

The Pituitary Gland

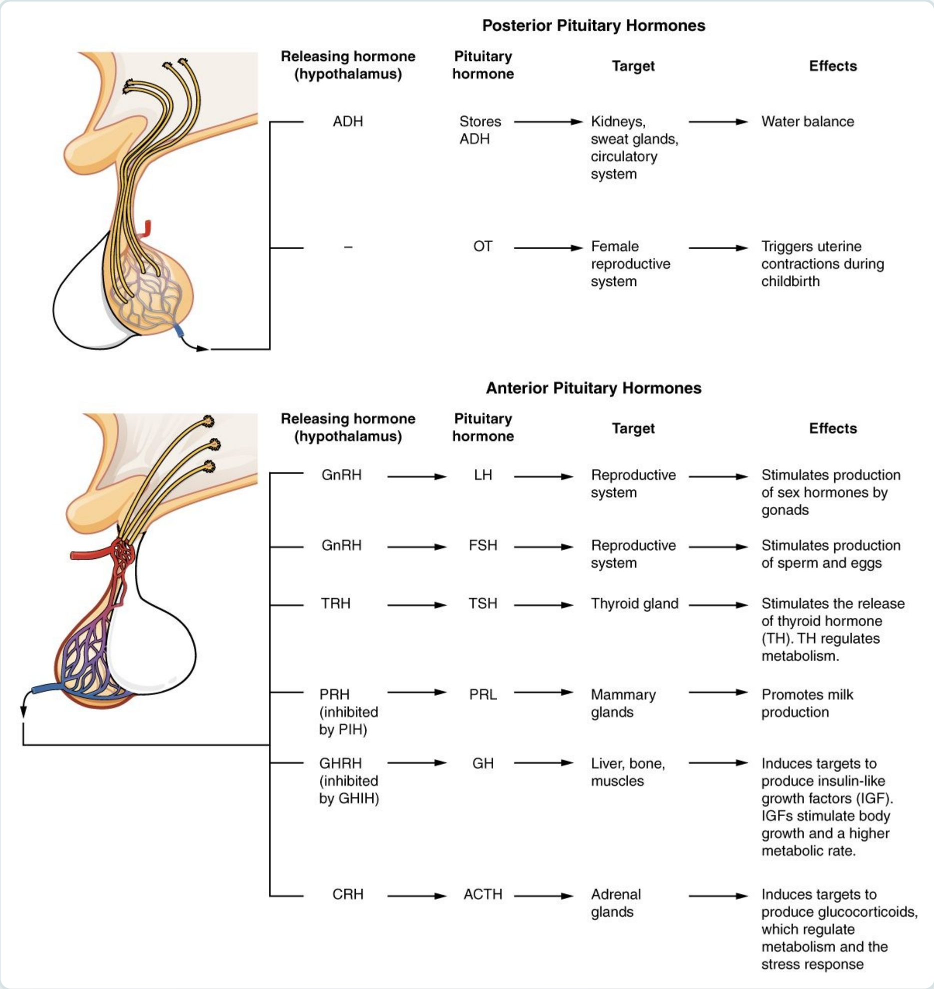


The pituitary gland and hypothalamus

- Sits beneath the hypothalamus
- Has anterior and posterior lobes
- Controlled by the hypothalamus
- Often called the "master gland"

SECTION 17.3

Pituitary Hormones



Pituitary hormones and their targets

- Anterior lobe makes tropic hormones
- Control the thyroid, adrenals, and gonads
- Plus growth hormone and prolactin
- Posterior releases ADH and oxytocin

17.4

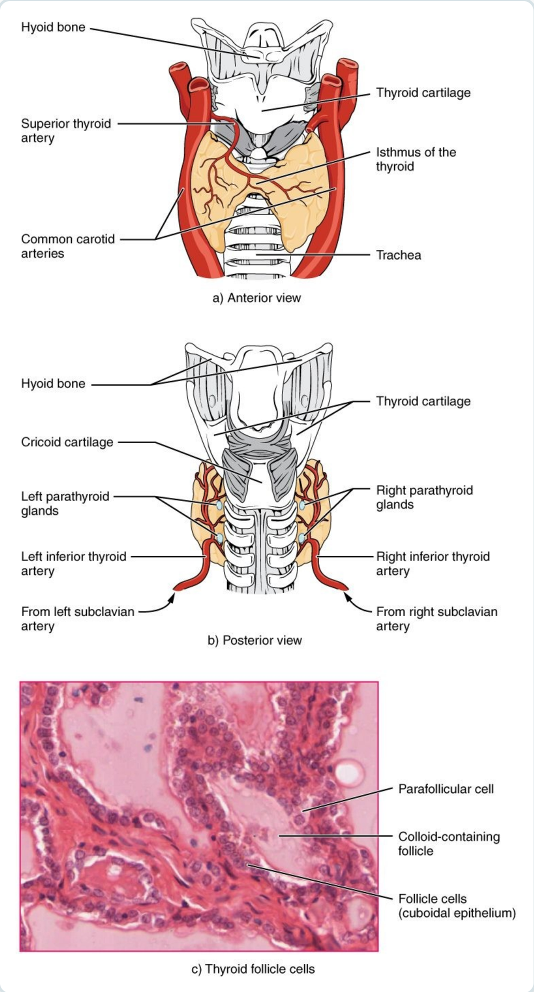
SECTION

The Thyroid Gland

The thyroid gland and its role in metabolism.

SECTION 17.4

The Thyroid Gland



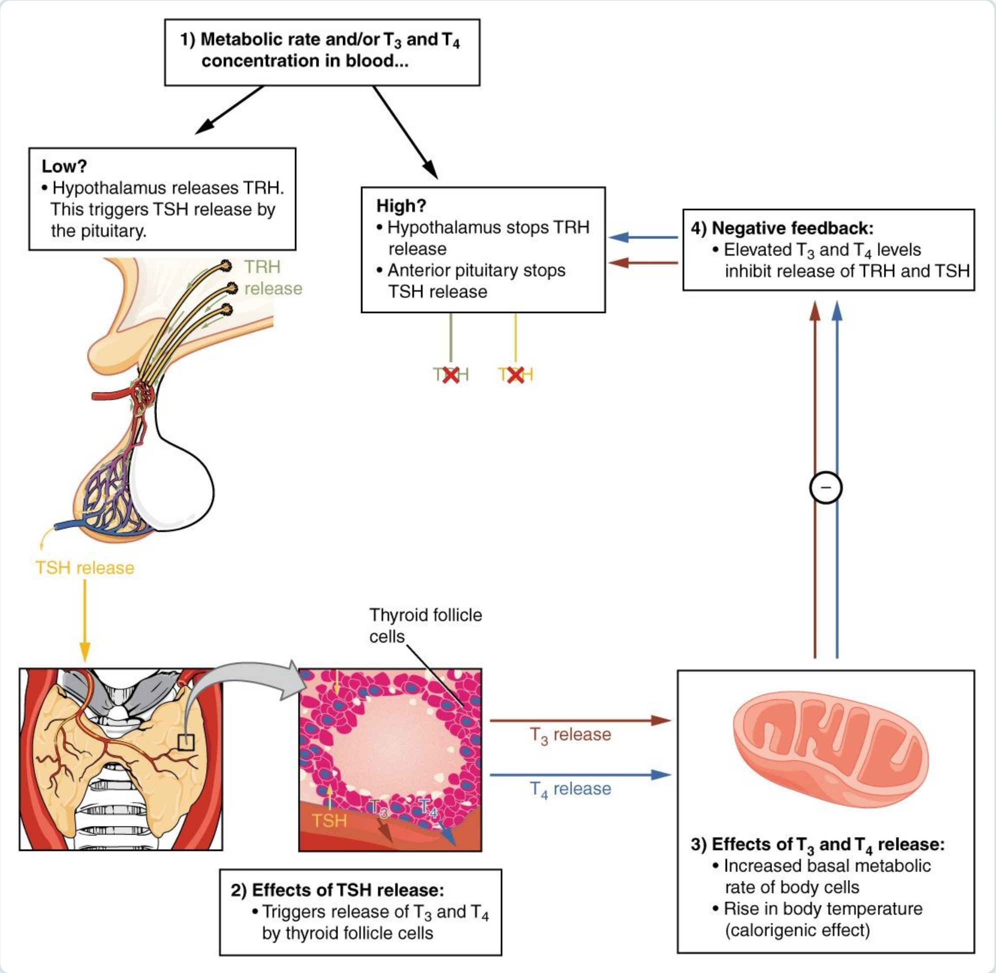
The thyroid gland

- Located in the front of the neck
- Wraps around the trachea
- Makes thyroid hormone (T3, T4)
- Requires iodine

SECTION 17.4

Thyroid Function & Feedback

- TSH from the pituitary drives it
- T3 and T4 raise metabolic rate
- Feedback keeps levels balanced
- Imbalance causes disease



Control of thyroid hormone

SECTION 17.4

Goiter

- An enlarged thyroid gland
- Often from iodine deficiency
- Can occur with over- or underactivity
- A visible neck swelling

Goiter (enlarged thyroid)

17.5

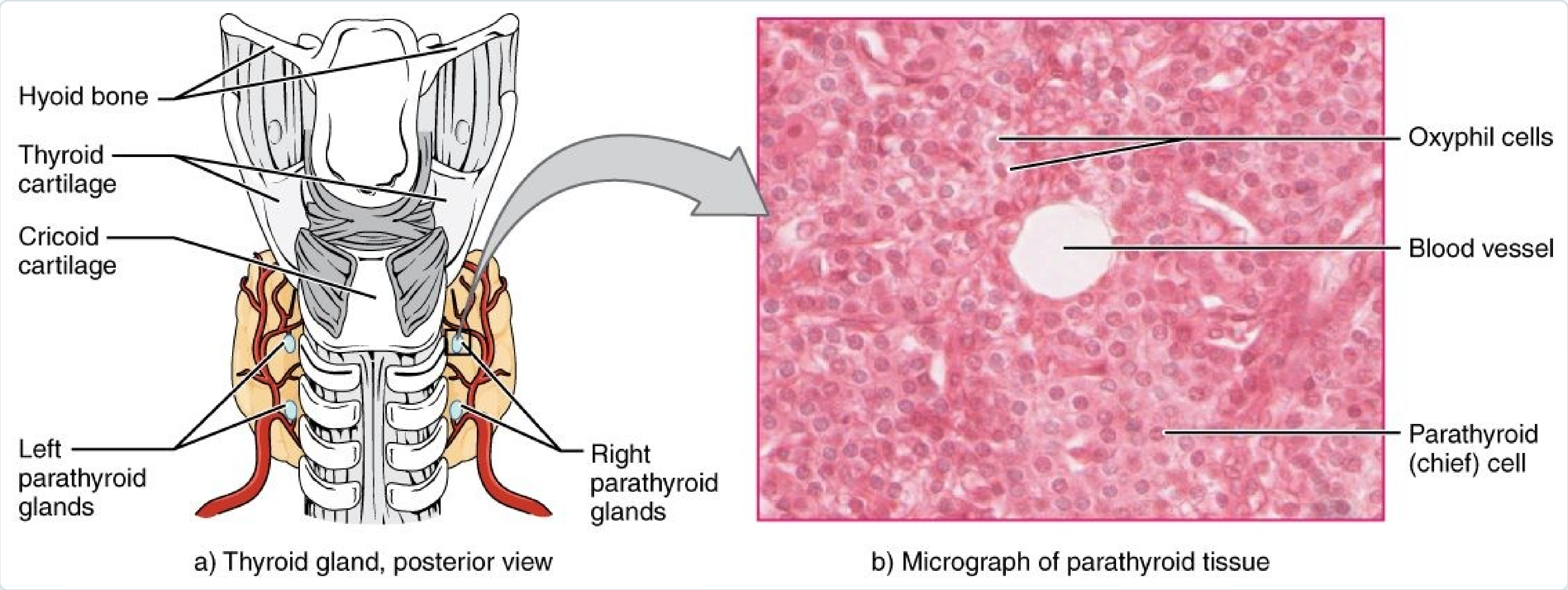
SECTION

The Parathyroid Glands

The parathyroid glands and calcium balance.

SECTION 17.5

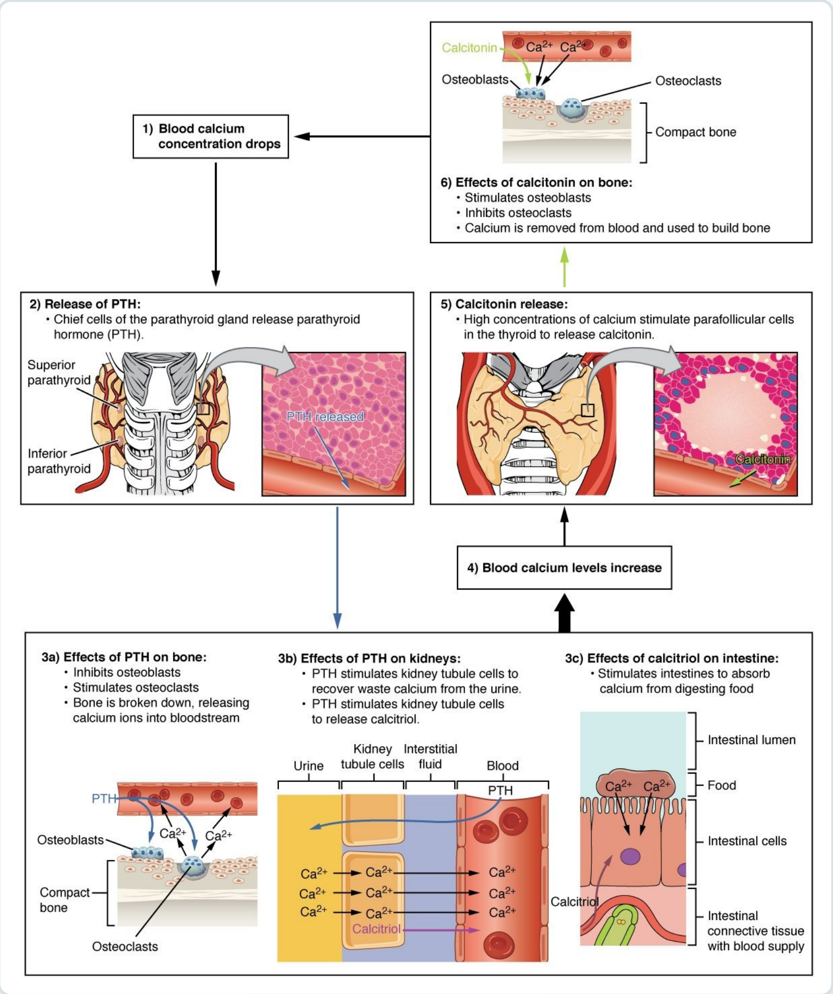
The Parathyroid Glands



The parathyroid glands

- Four small glands behind the thyroid
- Make parathyroid hormone (PTH)
- Tiny but essential
- Regulate blood calcium

Calcium Regulation



Hormonal control of blood calcium

- PTH raises blood calcium
- Acts on bone, kidney, and intestine
- Calcitonin lowers calcium
- Balance is tightly controlled

17.9

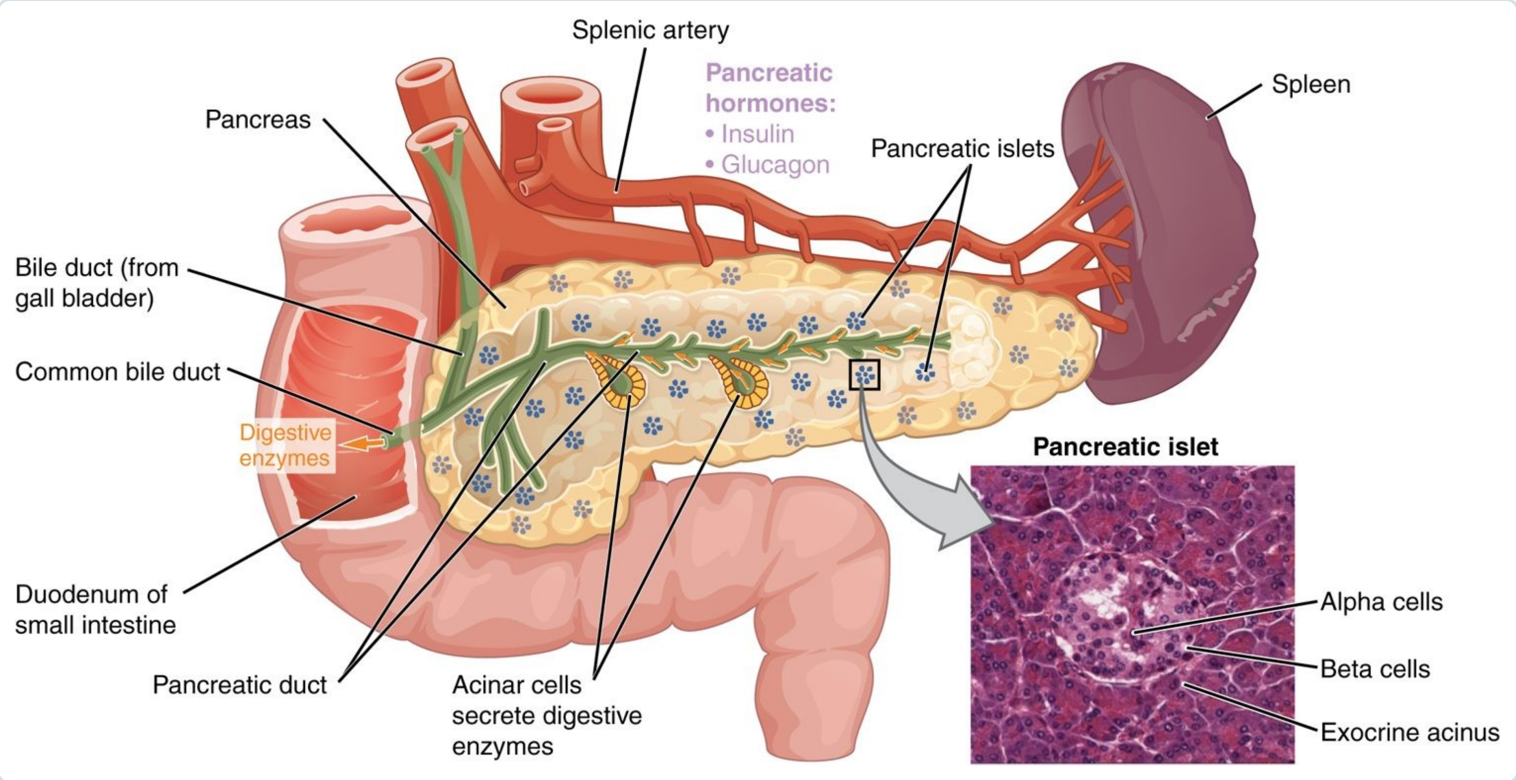
SECTION

The Endocrine Pancreas

The pancreas and the control of blood sugar.

SECTION 17.9

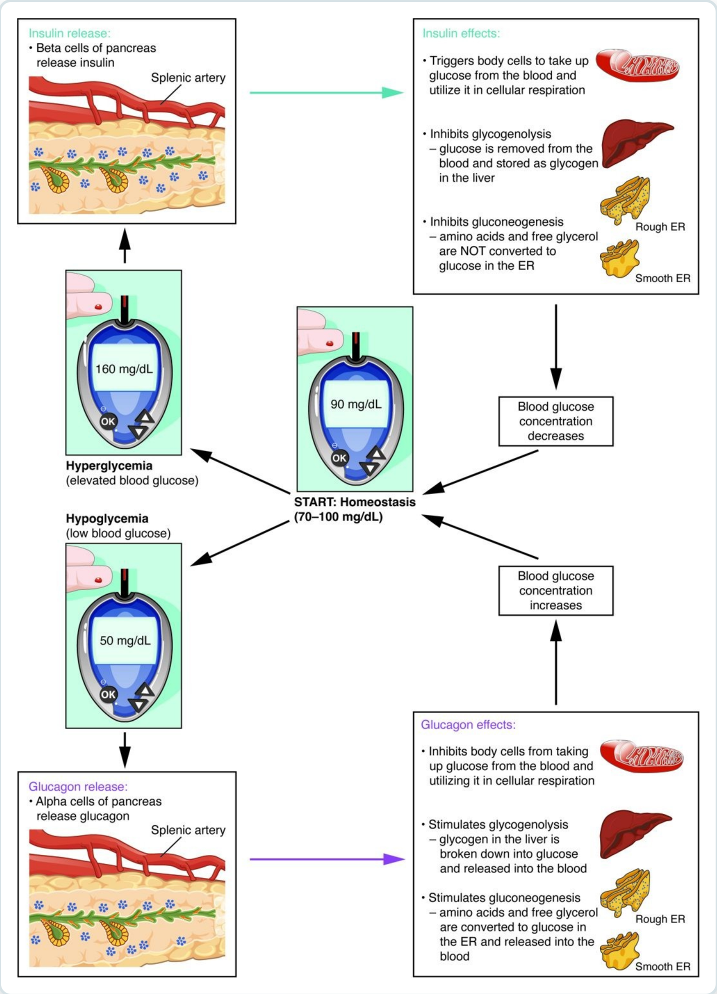
The Endocrine Pancreas



The endocrine pancreas and its islets

- Islets scattered in the pancreas
- Alpha cells make glucagon
- Beta cells make insulin
- Both regulate blood glucose

Blood Glucose Regulation



Insulin, glucagon, and blood glucose

- Insulin lowers blood glucose
- Glucagon raises it
- The two oppose to keep balance
- Failure causes diabetes

CHAPTER 17

Key Takeaways

- The endocrine system uses hormones in the blood to control slow, widespread body processes.
- Lipid-soluble hormones act on genes inside the cell; water-soluble ones use surface receptors.
- Negative feedback keeps most hormone levels in a stable range.
- The pituitary and hypothalamus direct the thyroid, adrenal glands, and gonads.
- The thyroid, parathyroids, and pancreas control metabolism, calcium, and blood sugar.

CLINICAL CONNECTIONS

Why the Endocrine System Matters in Practical Nursing

Diabetes Mellitus

Insulin problems disrupt blood sugar — a major focus of nursing care.

Thyroid Disorders

Hypo- and hyperthyroidism are common and very treatable.

Hormone Feedback

Many endocrine diseases stem from broken feedback loops.

Calcium Balance

Parathyroid problems disturb calcium and nerve/muscle function.

The Stress Response

Cortisol and the HPA axis shape illness and recovery.

Growth & Development

Pituitary hormones affect growth across the lifespan.